

- 10 Solve the following pairs of simultaneous equations.

a $y = 2x - 4$	b $5x - 2y = 14$
$3x + y = 11$	$2x + 2y = 14$
[by substitution]	[by elimination]

Review test 4 (Algebraic processes)

Allow 35 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- State the additive inverse of

a +8	b -23
c +664	d -0.37
- State the multiplicative inverse of

a 8	b $\frac{1}{10}$
c -12	d -0.8
- Remove brackets and simplify.

a $7x(x - 1) - 5(x - 1)$
b $a(2a + 5b) - b(a + b)$
- Expand the following.

a $(x + 8)(x - 2)$	b $(4x - 7)(3x + 2)$
--------------------	----------------------
- State the HCF of the following.

a $24xy$ and $8xy$	b $18xy^2$ and $12x^2y$
--------------------	-------------------------
- State the LCM of the following.

a $3a$ and $8b$	b $15xy$ and $2x^2y^2$
-----------------	------------------------
- Factorise the following.

a $5a + 5z$
b $2a^2 + 10a$
c $8m - 2 + 4nm - n$
d $6a + bm + 6b + am$
e $x^2 - 5x + 6$
f $4d^2 - 9e^2$

- 9 Solve the following.

a $4x > 20$
b $-11 < 3x$
c $x + 4 \geq 10x - 23$
d $2 - x < 14 - 3x$

- 10 Solve the following pairs of simultaneous equations.

a $y = x + 1$	b $5x + 3y = 1$
$x + y = 3$	$2x + 3y = -5$
[by substitution]	[by elimination]

Geometry and mensuration

- a Fig. P7 gives sketches and names of some common solids.

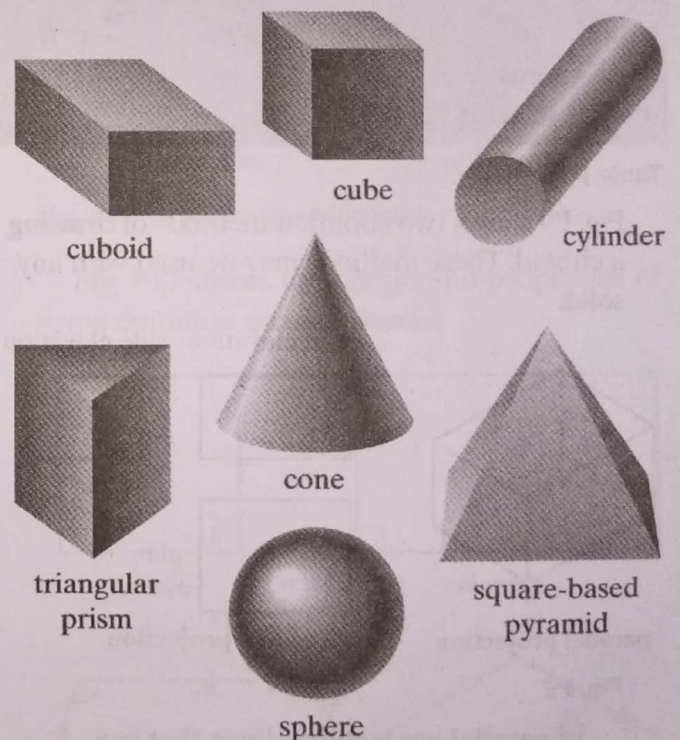


Fig. P7

All solids have **faces**; most solids have **edges** and **vertices** (Fig. P8).

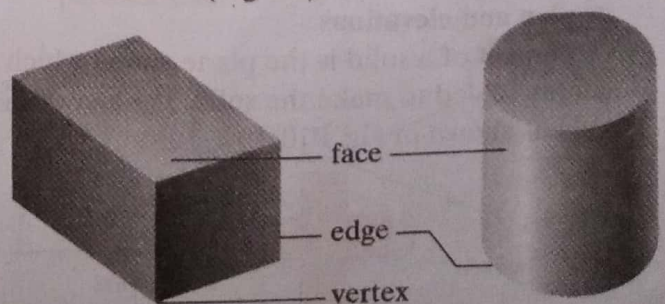


Fig. P8

Formulae for the **surface area** and **volume** of common solids are given in Table P1.

	surface area	volume
cube edge s	$6s^2$	s^3
cuboid length l , breadth b , height h	$2(lb + bh + lh)$	lbh
prism height h , base area A		Ah
sphere radius r	$4\pi r^2$	$\frac{4}{3}\pi r^3$
cylinder base radius r , height h	$2\pi r(r + h)$	$\pi r^2 h$
cone base radius r , height h , slant height l	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$

Table P1

Fig. P9 shows two common methods of **drawing** a cuboid. These methods may be used with any solid.

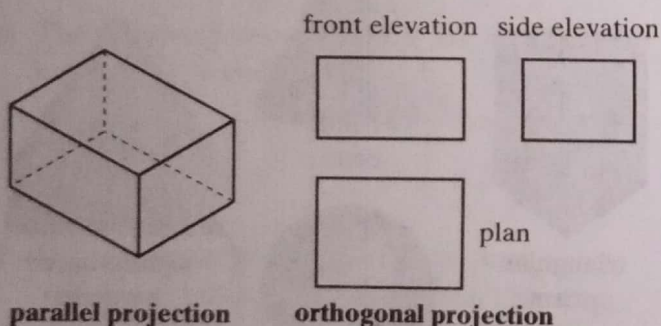


Fig. P9

In **parallel projections**, lines that are parallel on the solid appear parallel on the drawing. In **orthogonal projections**, the solid is represented by separate scale drawings of its **plan** and **elevations**.

The **net** of a solid is the plane shape which can be folded to make the solid. The net of a cube is shown in Fig. P10.



Fig. P10

b **Angle** is a measure of rotation or turning.

1 **revolution** = 360 **degrees** (1 rev = 360°)

1 **degree** = 60 **minutes** ($1^\circ = 60'$)

The names of angles change with their size. See Fig. P11.

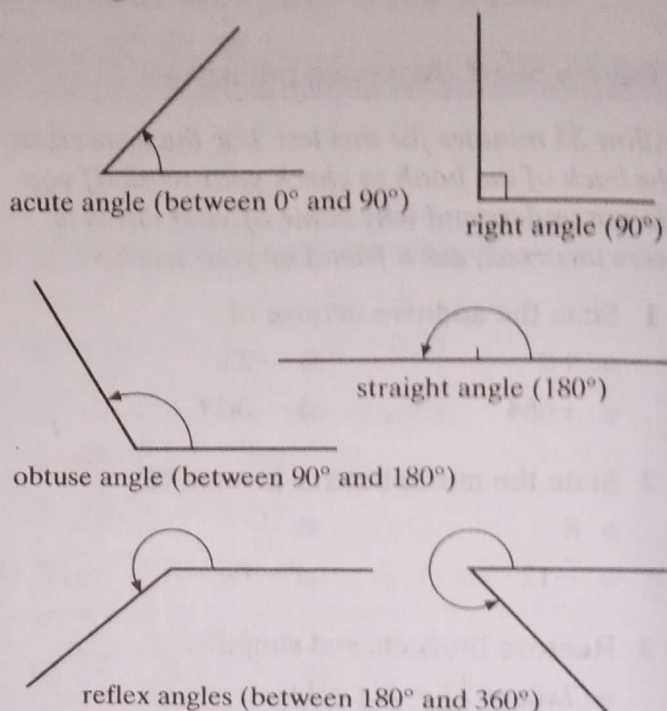
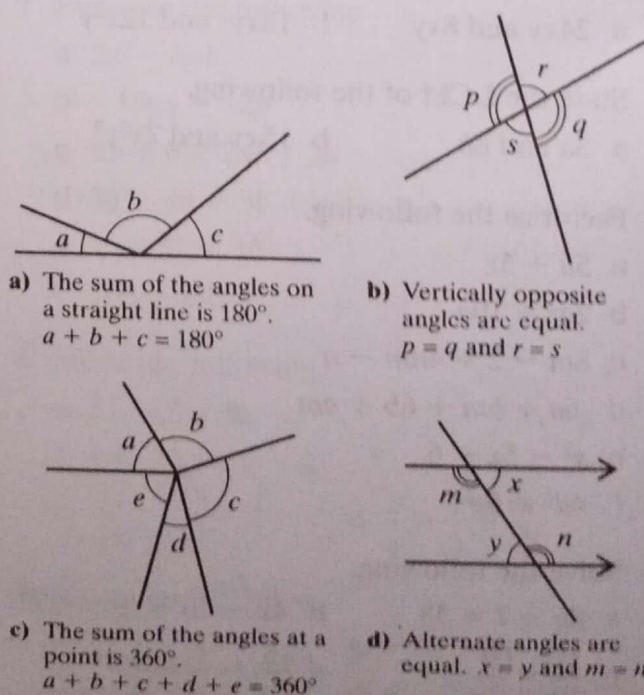


Fig. P11

Angles are measured and constructed using a **protractor**.

Fig. P12 shows some properties of angles formed when straight lines meet.



- e) Corresponding angles are equal. $a = b$ and $p = q$

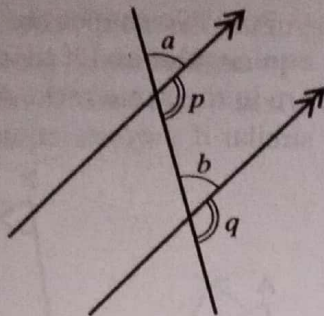


Fig. P12

In Fig. P13, α is the **angle of elevation** of the top of the flag-pole from the girl and β is the **angle of depression** of the girl from the boy.

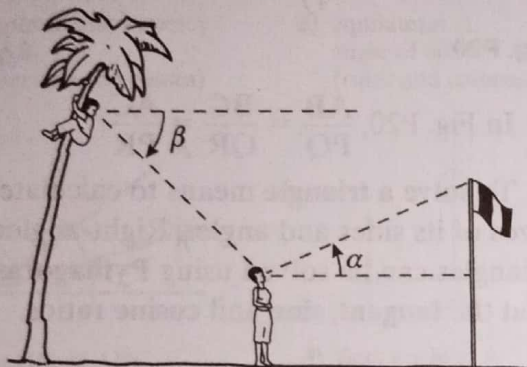


Fig. P13

Directions are taken from the **points of the compass** (Fig. P14).



Fig. P14

A **3-figure bearing** is a direction given as the number of degrees from north measured in a clockwise direction. See Fig. P15.

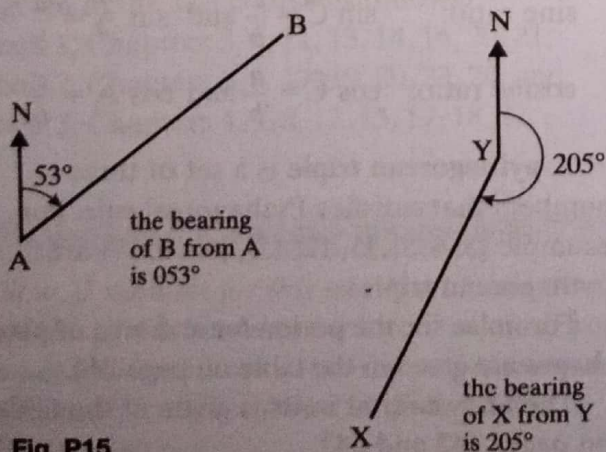


Fig. P15

The bearings in Fig. P15 can also be given as **compass bearings**: $N53^\circ E$ and $S25^\circ W$ respectively.

- c Fig. P16 shows the names and properties of some common **triangles**.

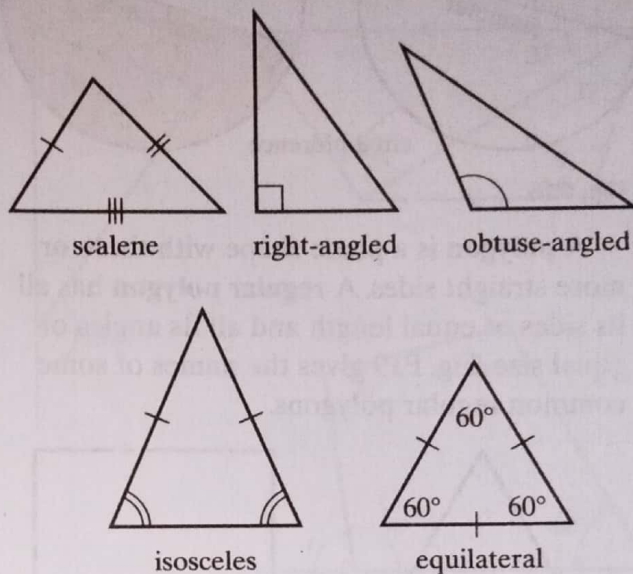


Fig. P16 Some common triangles

Fig. P17 shows the names and properties of some common **quadrilaterals**.

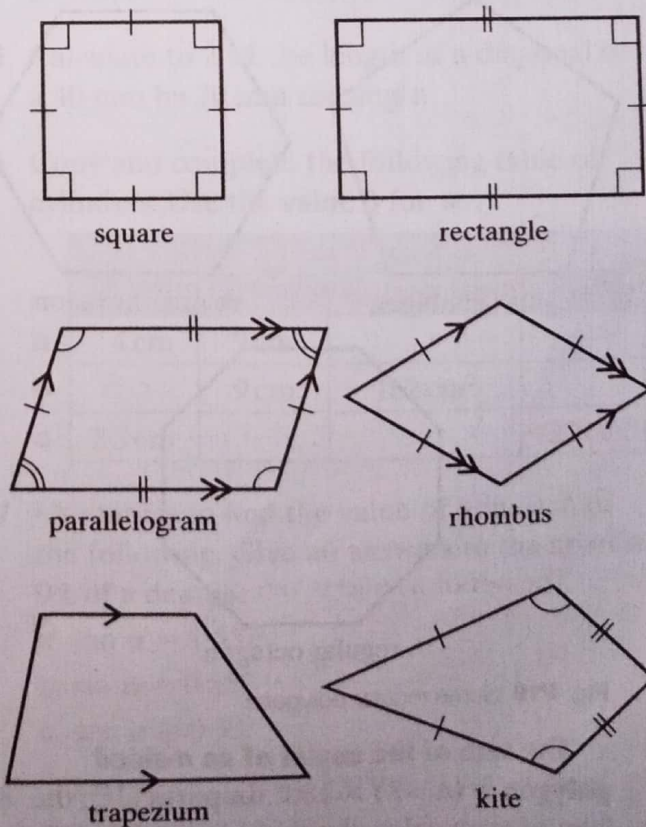


Fig. P17 Some common quadrilaterals

Fig. P18 gives the names of lines and regions in a **circle**.

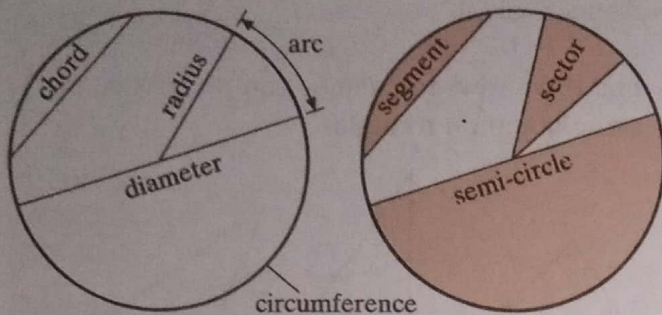
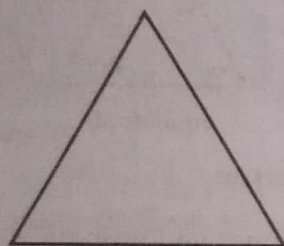
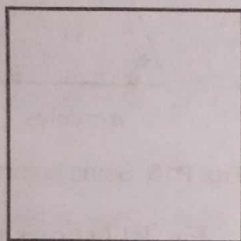


Fig. P18

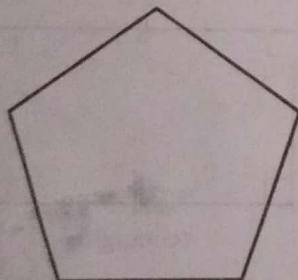
A **polygon** is a plane shape with three or more straight sides. A **regular polygon** has all its sides of equal length and all its angles of equal size. Fig. P19 gives the names of some common regular polygons.



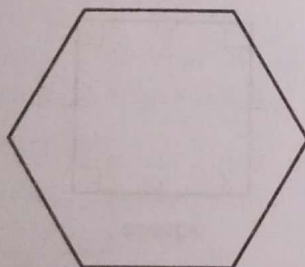
equilateral triangle



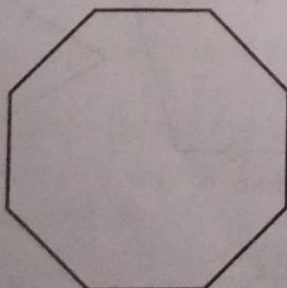
square



regular pentagon



regular hexagon



regular octagon

Fig. P19 Some regular polygons

The **sum of the angles of an n -sided polygon** is $(n - 2) \times 180^\circ$. In particular, the sum of the angles of a triangle is 180° and the sum of the angles of a quadrilateral is 360° .

Any two shapes are **similar** if they are **equiangular** and if their corresponding sides are in the same ratio. Any two triangles are similar if they are equiangular (Fig. P20).

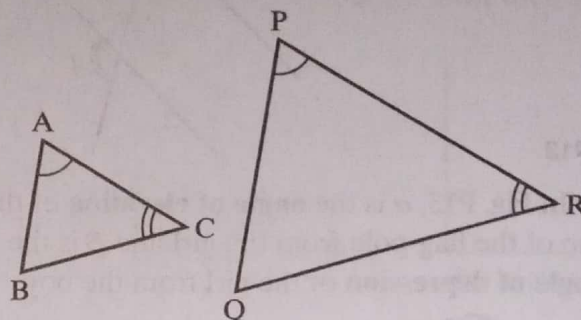


Fig. P20

$$\text{In Fig. P20, } \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$

To **solve a triangle** means to calculate the sizes of its sides and angles. Right-angled triangles can be solved using **Pythagoras' rule** and the **tangent, sine and cosine ratios**.

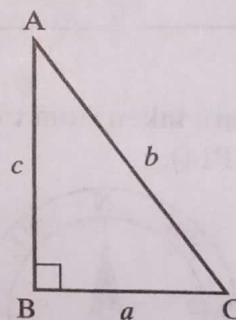


Fig. P21

In Fig. P21, $\triangle ABC$ is right-angled at B; side AC is the **hypotenuse**.

$$\text{Pythagoras' rule: } b^2 = a^2 + c^2$$

$$\text{tangent ratio: } \tan \hat{C} = \frac{c}{a} \text{ and } \tan \hat{A} = \frac{a}{c}$$

$$\text{sine ratio: } \sin \hat{C} = \frac{c}{b} \text{ and } \sin \hat{A} = \frac{a}{b}$$

$$\text{cosine ratio: } \cos \hat{C} = \frac{a}{b} \text{ and } \cos \hat{A} = \frac{c}{b}$$

A **pythagorean triple** is a set of three numbers that satisfies Pythagoras' rule. For example $\{3, 4, 5\}$, $\{5, 12, 13\}$, $\{8, 15, 17\}$ are pythagorean triples.

Formulae for the **perimeter** and **area of plane shapes** are given in the table on page 244.

The **SI system of units** is given in the tables on pages 242 and 243.

- d The sketches in Fig. P22 show the main features of the common **geometrical constructions**.

To construct angles of 45° and 30° , bisect angles of 90° and 60° respectively.

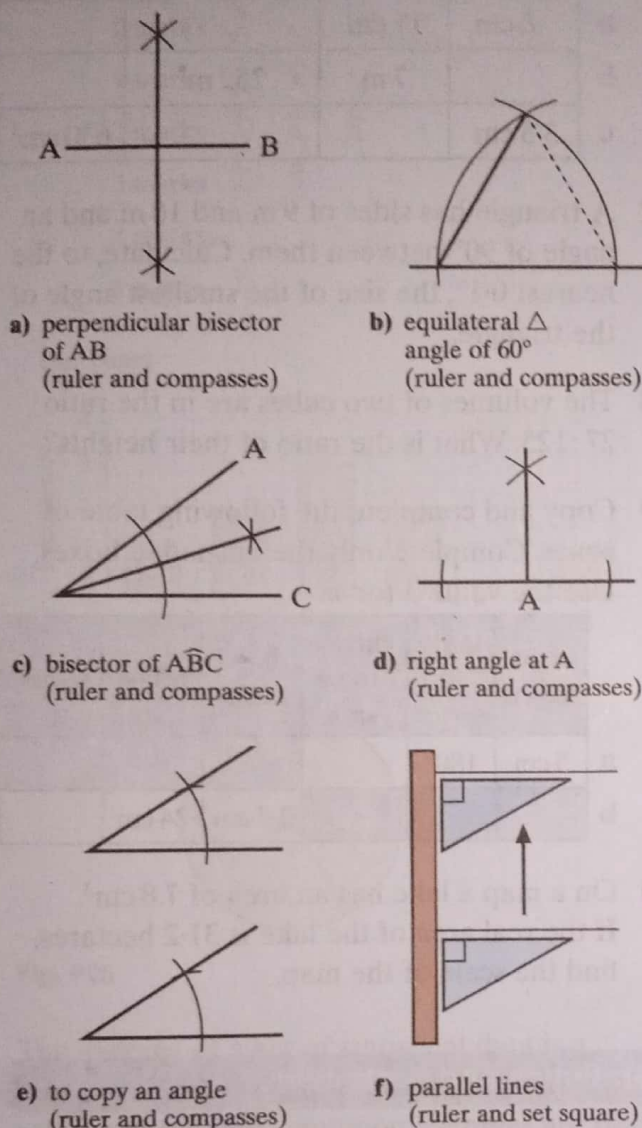


Fig. P22

Detailed coverage of geometry and mensuration is given in the JSS course as follows:

Book 1, Chapters 3, 6, 11, 13, 14, 16; 20, 21;
Book 2, Chapters 3, 6, 17, 19, 20, 23, 24; and
Book 3, Chapters 4, 7, 8, 12, 15, 17, 18, 22.

Review test 5 (Geometry and mensuration)

Allow 35 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 6.

- One of the angles of a parallelogram is 37° . What are the sizes of its other three angles?
- How many lines of bilateral symmetry does a square have?
- Find the values of m, n, p, q, r, s in Fig. P23.

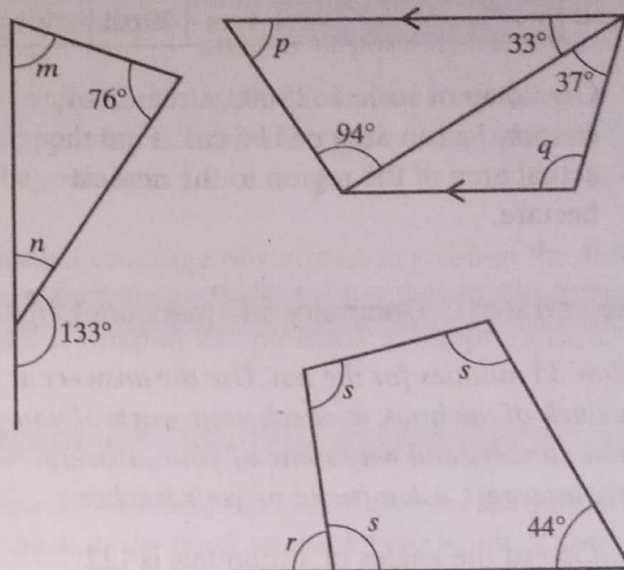


Fig. P23

- Which one of the following is a Pythagorean triple?
a 4, 5, 6 b 5, 13, 14 c 6, 8, 10
- Calculate to 2 s.f. the length of a diagonal of a 40 mm by 70 mm rectangle.
- Copy and complete the following table of cylinders. Use the value 3 for π .

	radius	height	curved surface area	volume
a	4 cm	9 cm		
b		9 cm	162 cm^2	
c	2.5 cm			120 cm^3

- Use tables to find the value of α in each of the following. Give all answers to the nearest 0.1 of a degree.
a $\tan \alpha = 1.552$
b $\sin \alpha = 0.555$
c $\cos \alpha = 0.33$
- $\triangle ABC$ is similar to $\triangle XYZ$. If $AC = 9 \text{ cm}$, $BC = 5 \text{ cm}$ and $YZ = 7 \text{ cm}$, find the length of XZ .

- 9 Copy the following table of cones. Complete only the unshaded boxes. Use the value 3 for π .

	slant height	angle of sector	curved surface area	base radius	height	volume
a	8 cm	210°				
b				4 cm	20 cm	

- 10 On a map of scale 1 : 25 000, a region of a country has an area of 11.5 cm². Find the actual area of the region to the nearest hectare.

Review test 6 (Geometry and mensuration)

Allow 35 minutes for the test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- One of the angles of a rhombus is 122°. What are the sizes of its other three angles?
- How many lines of bilateral symmetry does an equilateral triangle have?
- Find the values of p, q, r, s, t, x in Fig. P24.

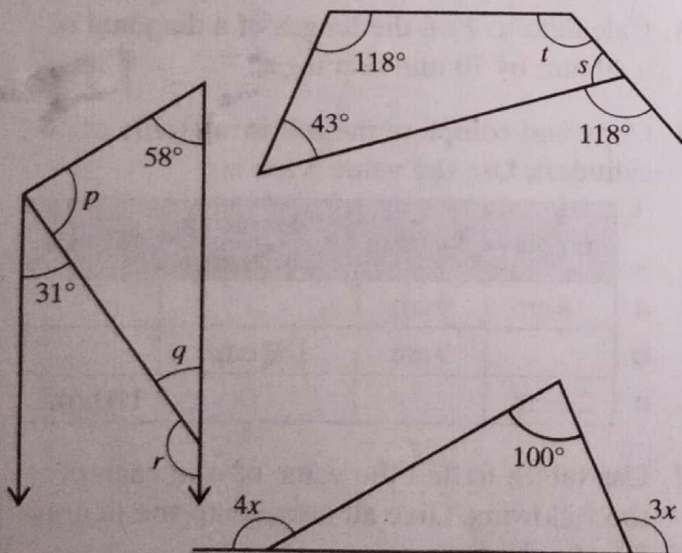


Fig. P24

- Two of the sides of a right-angled triangle are 3 m and 4 m. Find the size of the third side. [There are two possible answers.]
- A chord is 3 cm from the centre of a circle of radius 5 cm. What is the length of the chord?

- 6 Copy and complete the following table of cylinders. Use the value 3 for π .

	radius	height	curved surface area	volume
a	6 cm	11 cm		
b		7 m	252 m ²	
c	3.5 cm			630 cm ³

- A triangle has sides of 9 m and 10 m and an angle of 90° between them. Calculate, to the nearest 0.1°, the size of the smallest angle of the triangle.
- The volumes of two cubes are in the ratio 27 : 125. What is the ratio of their heights?
- Copy and complete the following table of cones. Complete only the unshaded boxes. Use the value 3 for π .

	slant height	angle of sector	curved surface area	base radius	height	volume
a	5 cm	180°				
b				2.5 cm	24 cm	

- 10 On a map a lake has an area of 7.8 cm². If the real area of the lake is 31.2 hectares, find the scale of the map.

Statistics and probability

- a Information in numerical form is called **statistics**. Statistical **data** may be given in **rank order** (i.e. in order of increasing size) such as in the following marks obtained in a test out of 5:

0, 1, 1, 2, 2, 2, 3, 3, 5

Data may also be given in a **frequency table** (Table P2).

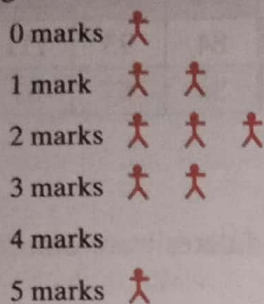
mark	0	1	2	3	4	5
frequency	1	2	3	2	0	1

Table P2

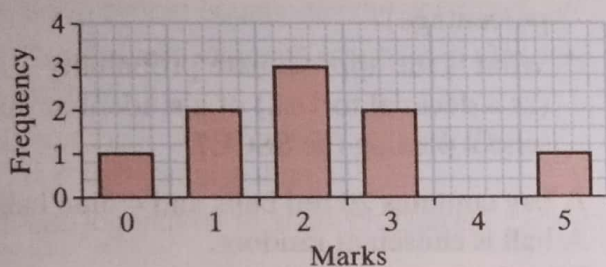
The **frequency** is the number of times each piece of data occurs.

Statistics can also be presented in graphical form. Fig. P25 shows the above data in a pictogram, a bar chart and a pie chart.

pictogram



bar chart



pie chart

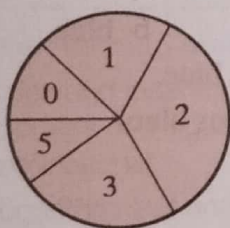


Fig. P25

- b The **average** of a set of statistical data is a number which is representative of the whole set. The three most common averages are the **mean**, the **median** and the **mode**. For the nine numbers given in rank order in paragraph a) above,

$$\text{mean} = \frac{0 + 1 + 1 + 2 + 2 + 2 + 3 + 3 + 5}{9} = 2\frac{1}{9}$$

The **median** is the middle number when the data are arranged in order of size (2).

Note: If there is an even number of data, the median is the mean of the middle two numbers.

The **mode** is the number with the greatest frequency (also 2 in this case). If a set of data has two modes we say it is **bimodal**.

- c **Probability** is a measure of the likelihood of a required outcome happening.

If an event *must* happen, probability = 1

If an event *cannot* happen, probability = 0

If the probability of something happening is p , then

$$\text{probability} = \frac{\text{number of required outcomes}}{\text{number of possible outcomes}}$$

$0 \leq p \leq 1$. If the probability of something happening is x , then the probability of it *not* happening is $1 - x$.

Detailed coverage of statistics is given in the JSS course as follows: Book 1, Chapters 17, 18, 22; Book 2, Chapter 21 and Book 3, Chapters 16, 23.

Review test 7 (Statistics and probability)

Allow 25 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 8.

- For each set of numbers, calculate the mean.
 - 13, 15, 2
 - 10, 11, 15
 - 0, 7, 5, 7, 7, 6
 - 9, 9, 5, 4, 13
- Find the mode, median and mean of each set of numbers.
 - 8, 8, 10, 13, 16
 - 10, 11, 14, 14, 15, 15, 15, 16, 16, 17
 - 8, 6, 3, 10, 6, 9
 - 2, 0, 6, 2, 7, 3, 1, 4, 2
- Table P3 shows the results of a survey of vehicles on a city road in 1 hour.

	cars	lorries	buses	others	total
number	26	9	9	6	

Table P3

- How many vehicles are there altogether?
- What is the probability that the next vehicle is a car?
- What is the probability that the next vehicle is *not* a car?
- What is the probability that the next vehicle is not a car, a lorry or a bus?

- 4 A number is chosen at random from the following set of numbers {31, 32, 33, ... 44, 45}. What is the probability that it is
- a a multiple of 4, b a prime number,
c the number 29, d a whole number?

Review test 8 (Statistics and probability)

Allow 20 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- 1 For each set of numbers, calculate the mean.
- a 23, 11, 13, 17
b 0, 8, 3, 5
c 3, 10, 0, 5, 6, 0
d 20, 21, 22, 23, 25, 27
- 2 Find the mode, median and mean of each set of numbers.
- a 14, 15, 15, 17, 18, 20
b 3, 7, 10, 10, 11, 11, 11
c 2, 0, 16, 0, 7, 11
d 9, 8, 6, 9, 6, 5, 7, 9, 7, 8, 9, 7

- 3 A school enters candidates for the SSCE examination. Table P4 shows the results for the years 2008–2011.

	2008	2009	2010	2011
candidates entered	84	93	111	112
number who passed	54	55	67	72

Table P4

During the 4 years:

- a How many candidates were entered for the SSCE?
b How many candidates passed?
c What is the school's success rate (as a percentage)?
d What is the approximate probability (as a decimal to 1 s.f.) of a student at this school passing the SSCE?
- 4 A bag contains 20 red balls and 4 blue balls. A ball is chosen at random. What is the probability that its colour is
- a red, b blue,
c either red or blue,
d neither red nor blue?

1

Numerical processes (1) Indices and logarithms

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Recall and use the laws of indices (multiplication, division, zero, reciprocal)
- 2 Simplify expressions that contain products of indices and fractional indices
- 3 Solve simple equations containing indices
- 4 Express and interpret numbers in standard form
- 5 Find the logarithms and antilogarithms of numbers greater than 1
- 6 Use logarithms to solve problems.

Teaching and learning materials

Teacher: Index and logarithm charts; graph chalkboard; books of 4-figure tables (as used in public examinations).

Student: Logarithm and antilogarithm tables (see pages 246 and 247 of this book)

This is a long chapter:

Indices and Standard Form Sections 1.1, 1.2 and 1.3

Logarithms Sections 1.4, 1.5 and 1.6

You may prefer to study Sections 1.1, 1.2 and 1.3 now and return to the rest of the chapter later.

1.1 Laws of indices (revision)

The following laws of indices are true for all values of a , b and $x \neq 0$.

- 1 $x^a \times x^b = x^{a+b}$
- 2 $x^a \div x^b = x^{a-b}$
- 3 $x^0 = 1, x \neq 0$
- 4 $x^{-a} = \frac{1}{x^a}$

Example 1

Simplify *a* $10^2 \times 10^3$ *b* $22n^7 \div 2n^3$ *c* 19^0
d 5^{-2} *e* $2^3 \times \left(\frac{1}{6}\right)^{-1}$ *f* $r \times r^0 \times r^{-5}$.

$$a \quad 10^2 \times 10^3 = 10^{2+3} = 10^5 \quad (\text{Law 1})$$

Check:

$$\begin{aligned} 10^2 \times 10^3 &= (10 \times 10) \times (10 \times 10 \times 10) \\ &= 10 \times 10 \times 10 \times 10 \times 10 \\ &= 10^5 \end{aligned}$$

$$b \quad 22n^7 \div 2n^3 = \frac{22}{2} \times n^{7-3} = 11n^4 \quad (\text{Law 2})$$

Check:

$$\begin{aligned} 22n^7 \div 2n^3 &= \frac{22 \times n \times n \times n \times n \times n \times n}{2 \times n \times n \times n} \\ &= 11 \times n \times n \times n \times n = 11 \times n^4 \\ &= 11n^4 \end{aligned}$$

$$c \quad 19^0 = 1 \quad (\text{Law 3})$$

Check:

$$\begin{aligned} 19^0 &= 19^{a-a} \quad (\text{since } a - a = 0) \\ &= \frac{19^a}{19^a} \quad (\text{Law 2}) \\ &= 1 \quad (\text{num. and denom. are equal}) \end{aligned}$$

$$d \quad 5^{-2} = \frac{1}{5^2} = \frac{1}{25} \quad (\text{Law 4})$$

Check:

$$\begin{aligned} 5^{-2} &= 5^{0-2} \\ &= \frac{5^0}{5^2} \quad (\text{Law 2}) \\ &= \frac{1}{5^2} \quad (\text{Law 3}) \\ &= \frac{1}{25} \end{aligned}$$

$$e \quad 2^3 \times \left(\frac{1}{6}\right)^{-1} = 8 \times \frac{1}{\frac{1}{6}} \quad (\text{Law 4})$$

$$= 8 \times 6 = 48$$

$$f \quad r \times r^0 \times r^{-5} = r^{1+0+(-5)} \quad (\text{Law 1})$$

$$= r^{-4} = \frac{1}{r^4}$$

$$\text{or } r \times r^0 \times r^{-5} = r^1 \times 1 \times \frac{1}{r^5} \quad (\text{Laws 3 and 4})$$

$$= \frac{1}{r^4}$$

Exercise 1a (Revision)

Simplify the following.

- | | |
|--|---|
| ① $10^5 \times 10^4$ | ② $a^3 \times a^9$ |
| ③ $5y \times 4y^4$ | ④ $2^2 \times 2^4$ |
| ⑤ $m^8 \div m^5$ | ⑥ $c^7 \div c$ |
| ⑦ $\frac{24x^6}{8x^4}$ | ⑧ $\frac{9 \times 10^9}{3 \times 10^3}$ |
| ⑨ 2^0 | ⑩ $6 \times z^0$ |
| ⑪ 4^{-3} | ⑫ $3x^{-3}$ |
| ⑬ $(1-4)^{-2}$ | ⑭ $(2-3)^{-1}$ |
| ⑮ $x^3 \div x^{-5}$ | ⑯ $a^{-9} \div b^0$ |
| ⑰ $(3x)^{-3}$ | ⑱ $9a^{-5} \times 4a^6$ |
| ⑲ $5x^2 \times 4x^0 \times 2x^{-6}$ | |
| ⑳ $15 \times 10^4 \div (3 \times 10^{-2})$ | |

1.2 Further laws of indices

Product of indices, $(x^a)^b$

Example 2

Simplify a $(x^2)^3$ b $(y^4)^2$ c $(z^3)^5$.

$$\begin{aligned} \text{a } (x^2)^3 &= x^2 \times x^2 \times x^2 \\ &= x^{2+2+2} = x^{2 \times 3} \\ &= x^6 \end{aligned}$$

$$\begin{aligned} \text{b } (y^4)^2 &= y^4 \times y^4 \\ &= y^{4 \times 2} \\ &= y^8 \end{aligned}$$

$$\begin{aligned} \text{c } (z^3)^5 &= z^3 \times z^3 \times z^3 \times z^3 \times z^3 \\ &= z^{3 \times 5} \\ &= z^{15} \end{aligned}$$

Notice in each part of Example 2, the final index is the **product** of the given indices, e.g. $(x^2)^3 = x^{2 \times 3} = x^6$.

In general, $(x^a)^b = x^a \times b = x^{ab}$.

Exercise 1b (Oral)

Simplify the following.

- | | | |
|-------------------|----------------|----------------|
| ① $(a^3)^2$ | ② $(b^2)^4$ | ③ $(c^5)^3$ |
| ④ $(d^4)^3$ | ⑤ $(e^3)^3$ | ⑥ $(f^0)^8$ |
| ⑦ $(g^{-2})^5$ | ⑧ $(h^4)^{-5}$ | ⑨ $(5^2)^{-1}$ |
| ⑩ $(3^{-2})^{-2}$ | ⑪ $(10^2)^7$ | ⑫ $(2^{-3})^2$ |

Example 3

Simplify the following.

a $(-3d^3)^2$ b $-3(d^3)^2$ c $(-4g^5)^3$ d $(a^3b)^4$

$$\begin{aligned} \text{a } (-3d^3)^2 &= (-3d^3) \times (-3d^3)^2 \\ &= -3 \times -3 \times d^3 \times d^3 \\ &= +9d^6 \end{aligned}$$

$$\begin{aligned} \text{b } -3(d^3)^2 &= -3 \times d^3 \times d^3 \\ &= -3d^6 \end{aligned}$$

$$\begin{aligned} \text{c } (-4g^5)^3 &= (-4g^5) \times (-4g^5) \times (-4g^5) \\ &= -64 \times g^{5 \times 3} \\ &= -64g^{15} \end{aligned}$$

or more quickly,

$$(-4g^5)^3 = (-4)^3 \times g^{5 \times 3} = -64g^{15}$$

$$\begin{aligned} \text{d } (a^3b)^4 &= (a^3b^1)^4 = a^{3 \times 4} \times b^{1 \times 4} \\ &= a^{12}b^4 \end{aligned}$$

Notice the following in Example 3:

- 1 The power outside the bracket raises everything inside the bracket to that power.
- 2 A negative number raised to an odd power is negative. A negative number raised to an even power is positive.

Exercise 1c

Simplify the following.

- | | |
|--------------------------------------|--------------------------------------|
| ① $(3m^4)^2$ | ② $(2n^5)^3$ |
| ③ $(4v^3)^2$ | ④ $4(v^3)^2$ |
| ⑤ $-2(a^2)^3$ | ⑥ $(-2b^2)^3$ |
| ⑦ $(-c^3)^2$ | ⑧ $(-e^4)^3$ |
| ⑨ $(-u^2)^5$ | ⑩ $-(c^5)^4$ |
| ⑪ $(-d^5)^4$ | ⑫ $(-f^4)^5$ |
| ⑬ $(mn^2)^4$ | ⑭ $(a^2b)^3$ |
| ⑮ $(x^2y^3)^4$ | ⑯ $(-u^3v^2)^4$ |
| ⑰ $(5mn^3)^3$ | ⑱ $(-4u^2v)^3$ |
| ⑲ $(-a^2m)^4$ | ⑳ $-3(de^3)^4$ |
| ㉑ $\frac{(-x^3)^2}{-x^4}$ | ㉒ $\frac{a^6}{(-a)^4}$ |
| ㉓ $\frac{(-c)^2 \times c^4}{(-c)^5}$ | ㉔ $\frac{-(d^2)^3}{d^4 \times (-d)}$ |

Fractional indices, $x^{\frac{1}{2}}$ and $x^{\frac{2}{3}}$

$\sqrt{x}$ is short for the square root of x .

$$\sqrt{x} \times \sqrt{x} = x$$

$$\text{Let } \sqrt{x} = x^p$$

$$\text{then, } x^p \times x^p = \sqrt{x} \times \sqrt{x} = x^1$$

$$x^{2p} = x^1$$

Therefore, $2p = 1$ (by equating indices)

$$\text{and } p = \frac{1}{2}$$

$$\text{Thus, } \sqrt{x} = x^{\frac{1}{2}}$$

Similarly, $\sqrt[3]{x}$ is short for the **cube root** of x . For example, $\sqrt[3]{8} = 2$ since $2 \times 2 \times 2 = 8$ and $\sqrt[3]{-27} = -3$ since $-3 \times -3 \times -3 = -27$.

$$\sqrt[3]{x} \times \sqrt[3]{x} \times \sqrt[3]{x} = x$$

$$\text{Let } \sqrt[3]{x} = x^q$$

$$\text{then, } x^q \times x^q \times x^q = x^1$$

$$x^{3q} = x^1$$

Therefore, $3q = 1$ (equating indices)

$$\text{and } q = \frac{1}{3}$$

$$\text{Thus, } \sqrt[3]{x} = x^{\frac{1}{3}}$$

$$x^{\frac{1}{2}} = \sqrt{x} \text{ and } x^{\frac{1}{3}} = \sqrt[3]{x}.$$

In general, $x^{\frac{1}{a}} = \sqrt[a]{x}$.

$$\text{Also } x^{\frac{2}{3}} = x^{2 \times \frac{1}{3}} = (x^2)^{\frac{1}{3}} = \sqrt[3]{x^2}$$

$$\text{or } x^{\frac{2}{3}} = x^{\frac{1}{3} \times 2} = (x^{\frac{1}{3}})^2 = (\sqrt[3]{x})^2$$

In general, $x^{\frac{a}{b}} = \sqrt[b]{x^a}$ or $(\sqrt[b]{x})^a$.

The laws of indices are summarised in Table 1.1.

$$1 \quad x^a \times x^b = x^{a+b}$$

$$2 \quad x^a \div x^b = x^{a-b}$$

$$3 \quad x^0 = 1$$

$$4 \quad x^{-a} = \frac{1}{x^a}$$

$$5 \quad (x^a)^b = x^{ab}$$

$$6 \quad x^{\frac{1}{2}} = \sqrt{x}$$

$$7 \quad x^{\frac{a}{b}} = \sqrt[b]{x^a} \text{ or } (\sqrt[b]{x})^a$$

where $x \neq 0$ in all cases

Table 1.1 Laws of indices

Example 4

Simplify a $9^{\frac{1}{2}}$ b $8^{\frac{1}{3}}$ c $8^{-\frac{2}{3}}$ d $4^{\frac{1}{2}} \times 4^{\frac{1}{3}}$

$$e \left(\frac{16}{81}\right)^{-\frac{3}{4}}$$

$$a \quad 9^{\frac{1}{2}} = \sqrt{9} = \pm 3$$

$$b \quad 8^{\frac{1}{3}} = \sqrt[3]{8} = 2$$

$$c \quad 8^{-\frac{2}{3}} = \frac{1}{8^{\frac{2}{3}}} = \frac{1}{(\sqrt[3]{8})^2} = \frac{1}{2^2} = \frac{1}{4}$$

$$d \quad 4^{\frac{1}{6}} \times 4^{\frac{1}{3}} = 4^{\frac{1}{6} + \frac{1}{3}} = 4^{\frac{1}{2}} = \sqrt{4} = \pm 2$$

$$e \quad \left(\frac{16}{81}\right)^{-\frac{3}{4}} = \frac{1}{\left(\frac{16}{81}\right)^{\frac{3}{4}}} = \left(\frac{81}{16}\right)^{\frac{3}{4}} = \left(\sqrt[4]{\frac{81}{16}}\right)^3$$

$$= \left(\frac{\pm 3}{\pm 2}\right)^3 = \pm \frac{27}{8}$$

Notice the following in Example 4.

- 1 The square root of a positive number can be positive or negative.
 $\sqrt{9} = \pm 3$. ± 3 means $+3$ or -3 .
- 2 $\sqrt[n]{n}$ means the n th root of n , i.e. the number which multiplied by itself n times gives n . For example $\sqrt[4]{16} = \pm 2$, since $2 \times 2 \times 2 \times 2 = 16$ and $-2 \times -2 \times -2 \times -2 = 16$.
- 3 A number raised to a negative power is equivalent to the reciprocal of the number raised to a positive power of the same numerical value.
For example, $\left(\frac{3}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$.

Example 5

Simplify a $\left(2\frac{1}{4}\right)^{1\frac{1}{2}}$ b $\left(\frac{8}{50}\right)^{-\frac{1}{2}}$ c $\sqrt{\frac{72a^3b^{-2}}{2a^5b^{-6}}}$

a Change mixed numbers to fractions.

$$\left(2\frac{1}{4}\right)^{1\frac{1}{2}} = \left(\frac{9}{4}\right)^{\frac{3}{2}} = \left(\sqrt{\frac{9}{4}}\right)^3 = \left(\pm\frac{3}{2}\right)^3 = \pm\frac{27}{8}$$

$$= \pm 3\frac{3}{8}$$

b Reduce the given fraction to its lowest terms.

$$\left(\frac{8}{50}\right)^{-\frac{1}{2}} = \left(\frac{4}{25}\right)^{-\frac{1}{2}} = \left(\frac{25}{4}\right)^{\frac{1}{2}} = \sqrt{\frac{25}{4}} = \pm\frac{5}{2} = \pm 2\frac{1}{2}$$